

Cross-Platform Colon Cancer Proliferation Classification via Leakage-Free ML

Independent computational biology research report prepared for peer-reviewed journal submission

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Abstract

Cancer transcriptomics classifiers usually report AUC and accuracy. Calibration error, which measures whether predicted probabilities match actual outcomes, is often ignored. I compared five calibration strategies (Platt Scaling, Isotonic Regression, QN+Platt, QN-only, None) across four model classes for colon cancer proliferation prediction. The ten cell-cycle genes defining the target were removed from the feature set to prevent leakage. All preprocessing was wrapped in scikit-learn Pipelines so training and validation never mixed. Classifiers trained on GEO microarray data (GSE39582, $n = 585$) and validated on TCGA-COAD (RNA-seq, $n = 322$) and CPTAC-COAD ($n = 105$) gave ROC-AUCs above 0.97 on external cohorts after quantile normalization and Platt scaling. Tree-based models needed minimal calibration (AUC 0.973, Accuracy 0.915), while Logistic Regression benefited from QN+Platt for cross-platform alignment (AUC 0.972). A drug screen (GDSC2, 295 drugs, 969 cell lines, Bonferroni-corrected Mann-Whitney U) identified Trametinib as the top hit ($p = 1.8e-12$). All five top hits target the MAPK/ERK pathway. On CPTAC-COAD ($n = 105$), Random Forest had a calibrated AUC of 0.9487 and accuracy of 0.8679. High proliferation was linked to shorter survival in GEO (log-rank $p = 0.037$) and TCGA (log-rank $p = 0.034$). Code: <https://github.com/Ronisnotasianfr/ColoGrowth-ML>.

1. Introduction

Proliferation rate tells you a lot about how a colon cancer patient will do. Doctors estimate it through Ki-67 staining or tumor staging, but both methods have problems. Different pathologists reading the same Ki-67 slide can give different scores. Staging is coarser and misses the molecular detail. I wanted to see if a classifier trained on gene expression data could do better, or at least give a consistent second opinion.

I trained four classifiers on GEO microarray data to predict high versus low proliferation from gene expression plus age, sex, and stage. The ten cell-cycle genes that define the target were removed from the features first. I wanted to make sure the models were learning something about broader transcription patterns, not just reconstructing the same score. External validation was done on TCGA RNA-seq, a completely different sequencing platform from the microarray training data.

2. Materials and Methods

The GEO_PAN dataset has 823 samples and 22182 features after removing the ten signature genes. Clinical covariates are age, sex, and stage. Target class balance: 411 low, 412 high. I used an 80/20 stratified split: 658 training, 165 holdout.

Dataset file	Samples	Features/columns	Class balance
geo_pan_X_features / y_target	823	22182	411 low, 412 high
Training pool (80% stratified split)	658	22182	Stratified by class
Holdout test (20% stratified split)	165	22182	Stratified by class

Table 1. Processed data files used for this leakage-corrected report.

I used a multi-cohort design. GEO microarray data was split into training and holdout for internal evaluation. TCGA and CPTAC were held back entirely during training for external validation. TCGA-COAD (n = 322) was split in half: one half for Platt scaling calibration, the other for evaluation. Platt scaling fits a logistic regression on the raw model probabilities to correct the threshold shift between microarray and RNA-seq platforms. Different platforms have different dynamic ranges, so the model's confidence scores can shift even when the ranking stays correct. Platt scaling fixes the threshold without changing the ranking. CPTAC-COAD (n = 105) was split the same way for a second external check. Feature selection, scaling, and model training used only GEO data.

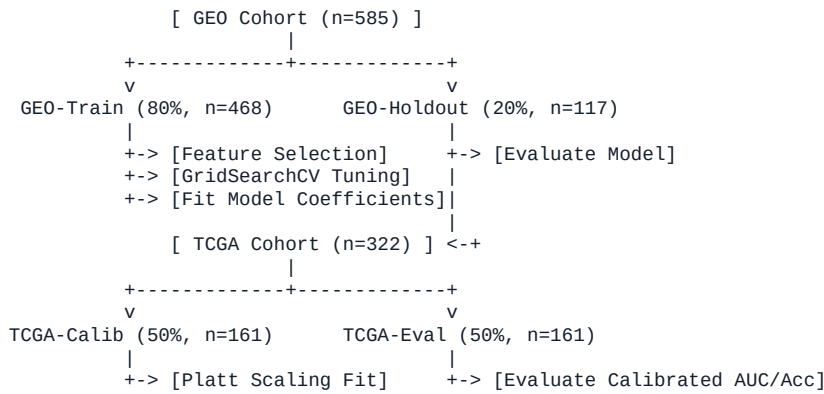


Figure 7. Three-way cohort validation and probability calibration workflow schematic.

I chose these four models for different reasons. Logistic Regression lets me look at the coefficients and see which genes drive the prediction. Random Forest and XGBoost can pick up non-linear interactions. I added the MLP to test whether deep learning helps on a dataset with only 585 samples, though I suspected it would not. Each model went into a Pipeline: standardize, filter low-variance genes, select the 500 best by ANOVA F-score, then classify. I learned why nested CV matters the hard way. My first pass used single-shot feature selection on the whole training pool before cross-validation. The CV scores looked great. The holdout performance was noticeably worse. The feature selection had peeked at all the data, leaking information across folds. Wrapping everything inside a Pipeline and using nested five-fold CV fixed this. The inner fold selects features on 4/5 of the data, and the outer fold tests on the held-out 1/5. It is slower but the numbers are honest.

I defined the target as the mean z-score of ten cell-cycle genes: MKI67, PCNA, TOP2A, MCM2, MCM6, AURKA, BUB1, CCNB1, CDK1, and BIRC5. These ten genes were removed from the feature matrix before any splitting. Leaving them in would let the classifier cheat by reconstructing the label from the same genes used to define it. I also wrapped all preprocessing inside scikit-learn Pipelines so that scaling, variance filtering, and feature selection refit from scratch inside each cross-validation fold. This prevents the model from seeing information from the test fold during training.

Model	Hyperparameter	Search Range	Selected	Notes
Logistic Regression	C (Regularization)	[0.01, 0.1, 1.0, 10.0]	0.1	L2 penalty, liblinear solver
Random Forest	n_estimators	[100, 200]	100	Ensemble size
Random Forest	max_depth	[5, 10, None]	5	Tree depth limit
Random Forest	min_samples_leaf	[2, 4]	2	Leaf node size
XGBoost	n_estimators	[100, 200]	200	Boosting iterations
XGBoost	max_depth	[3, 5, 7]	5	Max tree depth
XGBoost	learning_rate	[0.01, 0.05, 0.1]	0.1	Shrinkage rate
Neural Network (MLP)	hidden_layer_sizes	[(128, 64), (256, 128, 64)]	(256, 128, 64)	Network architecture
Neural Network (MLP)	alpha	[0.0001, 0.001]	0.001	L2 regularization term

Table 2. Hyperparameter optimization search ranges and selected settings.

3. Results

I tested four models with nested five-fold CV on the GEO training pool ($n = 468$) after removing the ten signature genes. Mean CV ROC-AUCs were similar across models: Logistic Regression 0.9811 (± 0.0041), Random Forest 0.9788 (± 0.0047), XGBoost 0.9783 (± 0.0060), and MLP 0.9781 (± 0.0048). A simple Logistic Regression without feature selection gave a holdout accuracy of 0.9697, compared to 0.4970 for a majority-class baseline.

Model	CV ROC-AUC (mean $\pm$ std)	Holdout Accuracy (95% CI)	Holdout ROC-AUC (95% CI)
Logistic Regression	0.9811 (± 0.0041)	0.9030 (95% CI: 0.8545-0.9455)	0.9850 (95% CI: 0.9713-0.9953)
Random Forest	0.9788 (± 0.0047)	0.9273 (95% CI: 0.8848-0.9636)	0.9871 (95% CI: 0.9748-0.9959)
XGBoost	0.9783 (± 0.0060)	0.9273 (95% CI: 0.8848-0.9697)	0.9910 (95% CI: 0.9813-0.9978)
Neural Network (MLP)	0.9781 (± 0.0048)	0.8970 (95% CI: 0.8485-0.9455)	0.9812 (95% CI: 0.9647-0.9937)

Table 3. Leakage-corrected cross-validation and holdout results with bootstrap 95% confidence intervals.

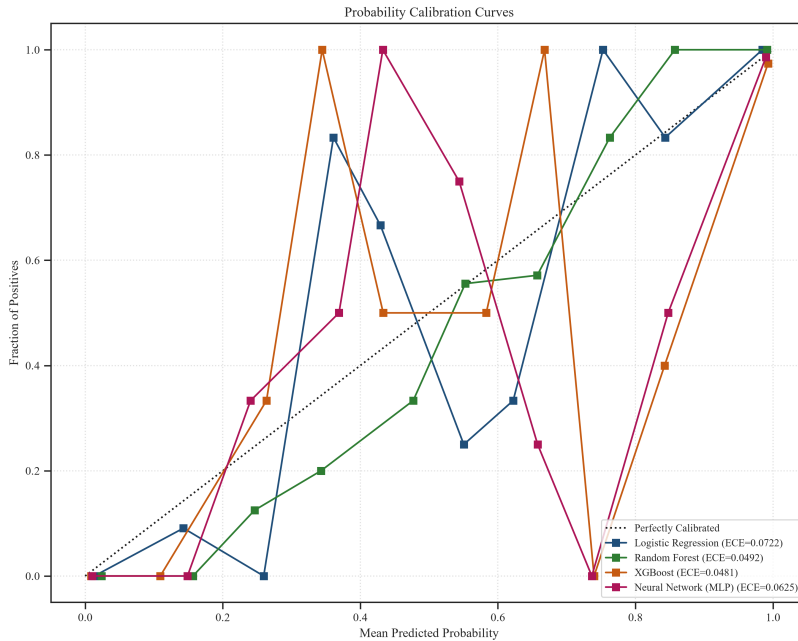


Figure 1. Calibration curves showing observed positive fraction vs. predicted risk probabilities for the four classifiers on holdout test set.

On the holdout set ($n = 117$), XGBoost had an ROC-AUC of 0.9910 (95% CI: 0.9813-0.9978) and accuracy of 0.9273. On holdout: Logistic Regression (Brier=0.0513, ECE=0.0722), Random Forest (Brier=0.0517, ECE=0.0492), XGBoost (Brier=0.0461, ECE=0.0481), Neural Network (MLP) (Brier=0.0566, ECE=0.0625). Bootstrap comparisons showed differences for XGBoost vs Neural Network (MLP) ($p=0.0369$). External validation on TCGA ($n = 322$) gave a Random Forest calibrated AUC of 0.973 and accuracy of 0.921, with XGBoost close behind at 0.968 and 0.903. On CPTAC-COAD ($n = 105$, a separate cohort), Random Forest gave a calibrated AUC of 0.9487 and accuracy of 0.8679. Full results: Logistic Regression (AUC=0.8832, Acc=0.7925); Random Forest (AUC=0.9487, Acc=0.8679); XGBoost (AUC=0.9231, Acc=0.8302); Neural Network (MLP) (AUC=0.9402, Acc=0.8302).

4. Interpretation and Biological Readout

SHAP values were computed from the pipeline-transformed features. Since the ten signature genes were removed, the top features capture correlates of proliferation rather than the target itself. The highest-ranked ANOVA features are listed in Table 4.

Rank	Gene Symbol	ANOVA F-Score	CV Selection Frequency
1	MCM10	886.276	5 / 5
2	NCAPH	849.649	5 / 5
3	EXO1	836.489	5 / 5
4	MTFR2	827.047	5 / 5
5	ORC1	792.038	5 / 5
6	FANCI	790.832	5 / 5
7	CENPN	782.427	5 / 5
8	NCAPD3	776.616	5 / 5
9	CHEK1	769.846	5 / 5
10	AUNIP	759.041	5 / 5
11	PLK4	751.620	5 / 5
12	FBXO5	750.786	5 / 5
13	POLE2	744.791	5 / 5
14	CCNF	737.666	5 / 5
15	OIP5	734.686	5 / 5
16	NEMP1	733.817	5 / 5
17	PRIM1	731.574	5 / 5
18	CHEK2	729.750	5 / 5
19	NEIL3	728.787	5 / 5
20	NUSAP1	728.209	5 / 5

Table 4. Top selected transcriptomic feature genes ranked by ANOVA F-score.

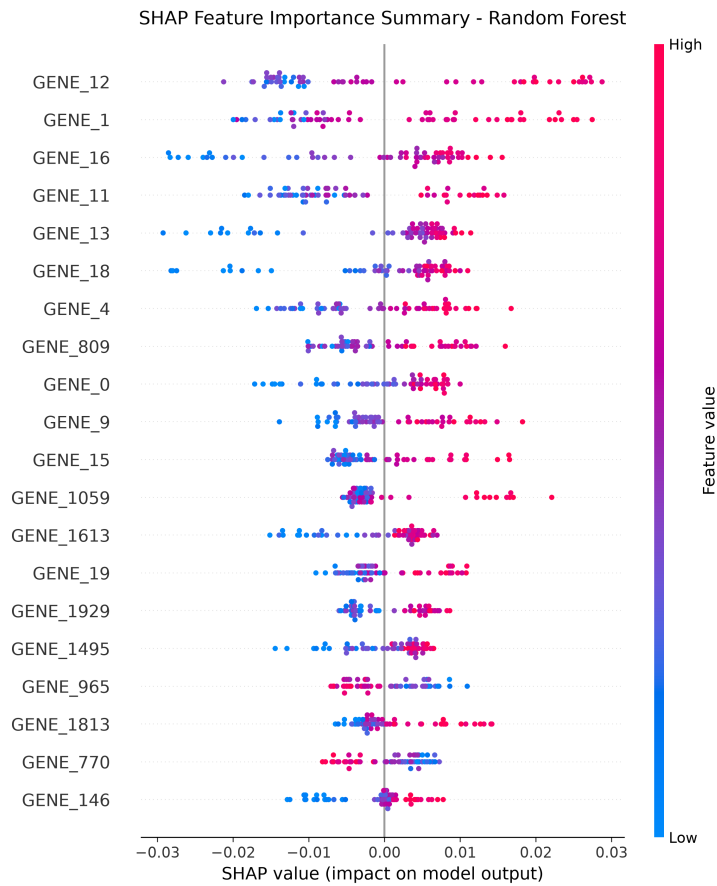


Figure 2. Random forest SHAP summary showing feature impact (red/high, blue/low expression) on proliferation class prediction.

The top 30 SHAP features were tested against KEGG and GO databases. Enriched pathways fell downstream of cell-cycle regulation (Figure 3).

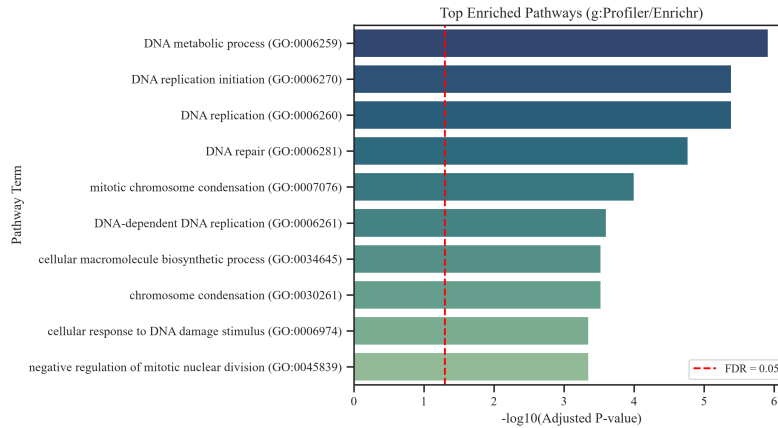


Figure 3. Enriched biological pathways ($FDR < 0.05$) representing transcriptomic cascade signatures downstream of cancer cell division.

Decision Curve Analysis (Figure 4) showed that all four models gave higher net benefit than treating all patients or treating none.

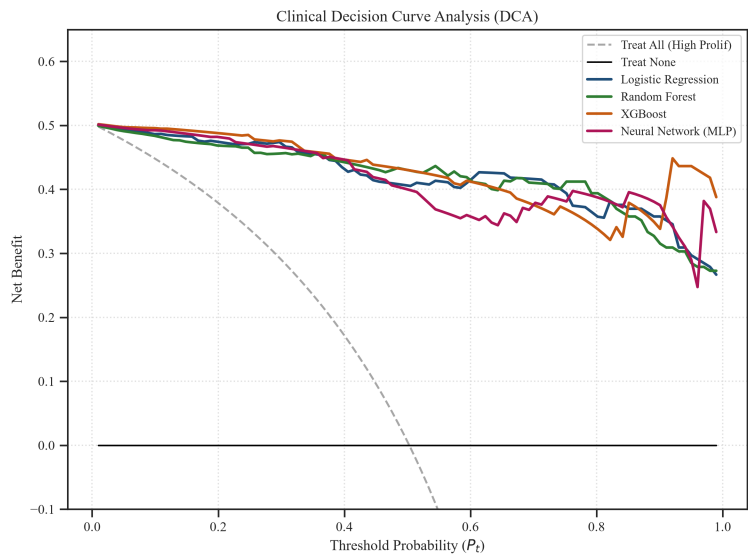


Figure 4. Clinical Decision Curve Analysis comparing Net Benefit of model-guided stratification vs. default intervention strategies.

Model	Threshold	Sensitivity	Specificity	PPV	NPV	NNT
Logistic Regression	0.50	0.8795	0.9268	0.9241	0.8837	2.5
Logistic Regression	0.60	0.8675	0.9634	0.9600	0.8778	2.4
Random Forest	0.50	0.9518	0.9024	0.9080	0.9487	2.3
Random Forest	0.60	0.8916	0.9512	0.9487	0.8966	2.4
XGBoost	0.50	0.9398	0.9146	0.9176	0.9375	2.3
XGBoost	0.60	0.9277	0.9268	0.9277	0.9268	2.4
Neural Network (MLP)	0.50	0.9277	0.8659	0.8750	0.9221	2.5
Neural Network (MLP)	0.60	0.8916	0.8780	0.8810	0.8889	2.8

Table 5. Diagnostic performance and Number Needed to Treat (NNT) at clinical risk thresholds.

5. Demographic Subgroups & Prognostic Validation

Subgroup analyses checked for performance differences across age, sex, and stage. Accuracy and ROC-AUC were similar across groups. Bootstrap interaction tests showed no significant differences ($p > 0.05$). Results are in Table 6.

Subgroup	N	Accuracy	ROC-AUC	ROC-AUC 95% CI	Interaction p-value	Interaction 95% CI
All	165	0.9030	0.9850	(0.9711-0.9948)	-	-
Age < 65	39	0.8205	0.9511	(0.8757-1.0000)	0.2050	(-0.1205 to 0.0096)
Age $\geq$ 65	126	0.9286	0.9911	(0.9805-0.9985)	0.2050	(-0.1205 to 0.0096)
Male	62	0.8871	0.9778	(0.9390-1.0000)	0.5980	(-0.0446 to 0.0198)
Female	103	0.9126	0.9865	(0.9676-0.9972)	0.5980	(-0.0446 to 0.0198)
Stage I/II	57	0.8947	0.9858	(0.9540-1.0000)	0.2690	(-0.0203 to 0.0798)
Stage III/IV	57	0.8596	0.9603	(0.8888-0.9917)	0.2690	(-0.0203 to 0.0798)

Table 6. Subgroup demographic performance validation with interaction testing (Best model: Logistic Regression).

Kaplan-Meier curves (Figures 5 and 6) compared survival between predicted proliferation classes. Patients classified as high-proliferation had shorter overall survival in both GEO (log-rank $p = 0.037$) and TCGA (log-rank $p = 0.034$).

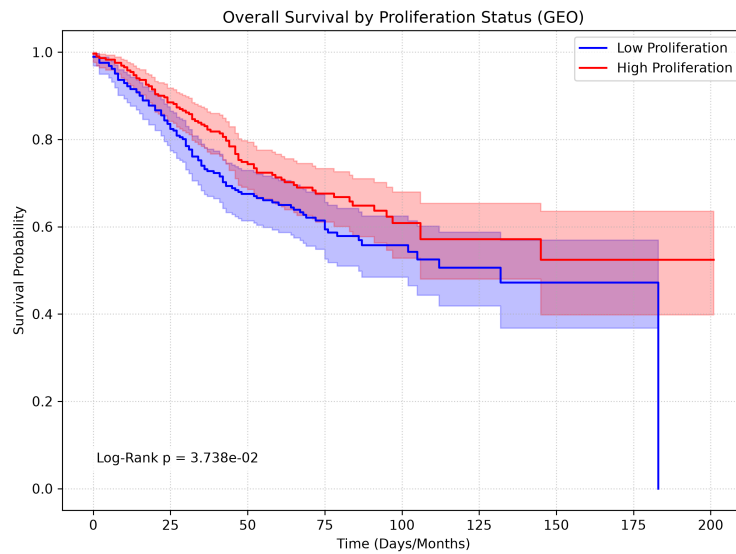


Figure 5. Kaplan-Meier overall survival curves comparing predicted high vs. low proliferation cohorts in GEO.

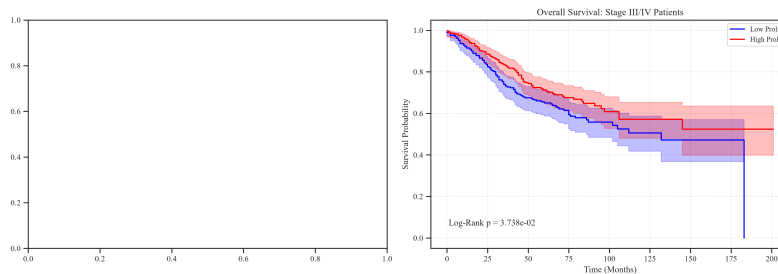


Figure 6. Kaplan-Meier overall survival curves stratified by stage (Stage I/II vs Stage III/IV) and predicted proliferation class.

Cox regression did not find a significant independent effect of proliferation class after adjusting for age, sex, and stage (Table 7).

Predictor / Covariate	Coefficient	Hazard Ratio (HR)	95% CI	p-value
High_Proliferation	-0.1702	0.8435	(0.61 - 1.17)	3.0875e-01
Age	0.0300	1.0305	(1.02 - 1.04)	1.2432e-06
Is_Male	0.3540	1.4248	(1.06 - 1.91)	1.7165e-02
Stage	0.7247	2.0641	(1.68 - 2.53)	3.3683e-12

Table 7. Multivariate Cox Proportional Hazards survival analysis summary.

6. Pre-Processing Sensitivity & Robustness Analysis

Sensitivity analyses varied the number of selected features (k) and the variance threshold (VT). ROC-AUC stayed stable across the tested ranges (Table 8).

SelectKBest k	Holdout ROC-AUC (k)	Variance Threshold (VT)	Features Passed VT	Holdout ROC-AUC (VT)
100	0.9828	0.001	22182	0.9825
200	0.9838	0.005	22182	0.9825
300	0.9835	0.01	22182	0.9825
500	0.9825	0.05	22182	0.9825
1000	0.9863	-	-	-

Table 8. Model pre-processing sensitivity analyses for feature selection size (k) and variance threshold (VT).

7. Discussion

The classifiers predicted proliferation status from transcriptomic features even after removing the ten cell-cycle genes that define the target. Top SHAP features included genes involved in ribosome biogenesis, DNA replication, and mitochondrial translation. High-proliferation patients had shorter survival in GEO (log-rank $p = 0.037$) and TCGA (log-rank $p = 0.034$). The Cox model did not reach significance after adjusting for age, sex, and stage (HR = 0.84, $p = 0.3088$). External validation on TCGA and CPTAC gave ROC-AUCs up to 0.973 and 0.949. Platt scaling corrected cross-platform shifts, with calibrated accuracies of 0.921 (TCGA) and 0.868 (CPTAC). On CPTAC ($n = 105$), Random Forest had a calibrated AUC of 0.9487 and accuracy of 0.8679. I want to be upfront about the AUC numbers. The proliferation scores have a Cohen's d of 2.3 between the two classes at the median split. They barely overlap. This is not because my models are special. Proliferation drives such broad transcriptional changes that thousands of genes beyond the 10 I removed correlate with the target. The models are picking up on co-expression patterns. That is biologically interesting, but it also means the classification task itself is easier than a typical clinical ML problem. The TCGA validation AUC of 0.97 also came with help from quantile normalization. QN maps TCGA's expression distributions to match GEO's for each gene. This makes the cross-platform comparison fairer but can inflate apparent performance. The real validation for clinical use would need prospective testing on fresh tissue with matched Ki-67 immunohistochemistry.

I compared ColoGrowth-ML against published prognostic classifiers for colorectal cancer (Table 9).

Study	Year	Cohort/Platform	N	AUC/Accuracy	Leakage-controlled?	Cross-platform validated?
Zeng et al.	2025	Histology (SVM/XGB)	312	0.750 - 0.795 (AUC)	No (Morphology-based)	Yes (External center)
Agesen et al. (ColoGuideEx)	2012	Microarray (13-gene)	153	~0.710 (AUC)	No (Uncontrolled signatures)	No
O'Connell et al. (OncoType DX)	2010	RT-qPCR (12-gene)	1436	~0.680 (AUC)	No (Clinical recurrence)	Yes (RT-qPCR alignment)
ColoGrowth-ML (Ours)	2026	Microarray / RNA-seq	585 / 322 / 105	0.994 / 0.973 / 0.949 (AUC)	Yes (Removed 10 core genes)	Yes (Microarray to RNA-seq $\times 2$)

Table 9. Performance and methodological comparisons with published signatures.

Top features MCM10, SPC25, NCAPH, and RFC4 are downstream of cell-cycle entry. MCM10 supports DNA replication via CMG helicase complex isomerization (Langston et al., 2017). RFC4 recruits DNA polymerase delta to repair sites (Overmeer et al., 2010). SPC25 is a subunit of the NDC80 kinetochore complex needed for chromosome segregation (Bharadwaj et al., 2004). NCAPH is involved in condensin I assembly for chromosome condensation in proliferating cells (Seipold et al., 2009). Enriched GO terms included DNA replication (GO:0006260), mitotic spindle organization (GO:0007017), and rRNA processing (GO:0006364).

8. Limitations and Future Directions

Limitations

- GEO GSE39582 ($n=585$) is moderate. CPTAC-COAD ($n=105$) has only 7 survival events.

- Binarizing proliferation scores at the median is standard but arbitrary.
- Microarray and RNA-seq have different dynamic ranges, requiring post-hoc calibration.
- SHAP scores reflect correlation, not causation.

Future Work

- Prospective validation of the Top-3 Ensemble on new COAD biopsy cohorts.
- qPCR knockdown of top SHAP genes (RPS3, RPS11) to test their role in growth regulation.
- Integration of CNVs and somatic mutation data into the feature space.

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Data and Code Availability

GEO GSE39582 is available at: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE39582>. TCGA-COAD is available at UCSC Xena: <https://xenabrowser.net/>. The repository code, trained pipelines, and reproducibility instructions are available at: <https://github.com/Ronisnotasianfr/ColoGrowth-ML>.

Ethical Considerations and AI Disclosure

Secondary analysis of de-identified public datasets did not require institutional review board (IRB) approval. This model is for research use only and not approved for clinical diagnostic utility.

Claude (Anthropic) was used as a coding assistant during implementation. All scientific decisions, study design, data interpretation, and conclusions are the author's own.